

Damghan University

Master's Thesis

Computer Science (Soft Computing and Artificial Intelligence)

Development of a Hybrid Method Based on Particle Swarm Optimization and Genetic Algorithm for Selecting an Effective Feature Subset

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Abstract

With the increasing volume of data and the growing number of features in machine learning problems, feature selection has become one of the most important stages of data preprocessing. The presence of irrelevant and redundant features can increase computational complexity, reduce processing speed, and decrease the performance of classification models. Therefore, the use of optimization-based methods for selecting effective features plays a significant role in improving the accuracy and efficiency of machine learning models. In this study, a hybrid method based on the PSO and GA metaheuristic algorithms is proposed for optimal feature selection, and its performance is evaluated using the Random Forest classifier. In the proposed approach, by simultaneously utilizing the global search capability of the PSO algorithm and the ability of the Genetic Algorithm to generate diverse solutions, an optimal subset of effective features was selected. These selected features were then used as inputs for the Random Forest classification model. To evaluate the proposed method, the Spambase and Phishing datasets were employed. The results showed that in the Spambase dataset, 46 features were selected from the initial 57 features, and the model achieved an accuracy of 96.79%, precision of 96.40%, recall of 97.89%, and an F1-score of 97.14%. In addition, the AUC value of 0.988 indicated the high capability of the model in distinguishing between classes. In the Phishing dataset, 22 features were selected from the initial 30 features, and the classification results included an accuracy of 98.20%, precision of 97.98%, recall of 98.80%, and an F1-score of 98.39%. Furthermore, the AUC value of 0.996 demonstrated the strong discriminative power of the selected features in separating classes. Overall, the results of this research indicate that the hybrid PSO–GA method is an effective approach for feature selection in classification problems and can improve model performance, reduce computational complexity, and increase processing speed while simultaneously decreasing the number of features.

Keywords: Feature Selection, Genetic Algorithm, Particle Swarm Optimization, Machine Learning, Random Forest, Classification.

Problem Statement

With the increase in large and complex data in various problems, the presence of a large number of features in the data has become one of the main challenges in machine learning and analysis. These features can include textual, numerical, structural or behavioral information, each of which may have a different contribution to the prediction models. Many of these features are unimportant, repetitive or noisy, and their use can reduce the accuracy of the models, increase the error rate and increase the computational complexity. In addition, the presence of ineffective features leads to an increase in the dimensions of the data and, as a result, longer model training time, higher consumption of computational resources and a decrease in the overall efficiency of the learning process. Therefore, the optimal and targeted selection of effective features, as a crucial preprocessing step, plays a key role in improving the performance of machine learning models and can help simplify models, reduce computational cost, and increase prediction accuracy [4,5]. The feature selection problem can be considered as a multi-objective optimization problem, such that the goal is to simultaneously maximize prediction accuracy and minimize the number of features used. In practice, the search space of all possible combinations of features is very large and complex, and it is impossible to explore all options. This has led researchers to use intelligent methods and population-based algorithms for feature selection, because these algorithms have the ability to efficiently search in a large feature space and find optimal subsets [6].

One of the most widely used algorithms in this field is the particle swarm optimization algorithm, which has a high ability to find suitable feature sets by utilizing group and simultaneous particle search in the solution space. However, the particle swarm optimization algorithm may sometimes get stuck in local optima and fail to achieve the best answer in the search space. On the other hand, the genetic algorithm increases the search diversity by using combination

and mutation operators and increases the chance of escaping from local optima, but its convergence speed is usually lower and it requires more time to reach the optimal answer. By combining these two algorithms, the advantages of both methods can be combined: the high search speed of the particle swarm optimization algorithm and the ability of the genetic algorithm to maintain diversity and prevent getting stuck in local optima. This combination allows finding a subset of features that has the greatest impact on improving the accuracy of learning models while reducing computational complexity. Using such a hybrid algorithm allows researchers to perform feature selection intelligently and efficiently and improve the performance of models in problems with large and complex data.

The main goal of this research is to develop an intelligent and optimal method for feature selection using a combination of particle swarm optimization and genetic algorithms that can identify an effective subset of features, increase the accuracy of learning models, and at the same time reduce computational complexity and processing time. This method can be used as a general feature selection framework in various machine learning problems and has high generalizability.

Importance of the research

With the increase in the volume and complexity of data in various machine learning problems, the presence of a large number of features has become one of the main challenges in data analysis. Many of these features are unimportant, repetitive, or noisy, and their use can reduce the accuracy of models, increase the error rate, and increase computational complexity. Therefore, effective feature selection plays a key role in improving accuracy, reducing data volume, and increasing the efficiency of models.

The feature selection problem can be defined as a multi-objective optimization problem, in which the simultaneous goal is to maximize the prediction accuracy and minimize the number of features used. Due to the large dimensions of the search space, it is practically impossible to explore all possible combinations of features [7].

To solve this problem, the use of intelligent and population algorithms is proposed. The particle swarm optimization algorithm is able to find effective subsets of features by group searching in the solution space, but it may get stuck in local optima. In contrast, the genetic algorithm increases the search diversity by using combination and mutation operators, but its convergence speed is usually slower. The combination of these two algorithms can prevent getting stuck in local optima and improve the feature selection process while maintaining the search speed.

Research Objectives

1- Developing an optimal method for selecting effective features:

Design and implement a hybrid algorithm based on particle swarm optimization and genetic algorithm (PSO-GA) to identify an optimal subset of features that have the greatest impact on improving the accuracy of learning models. This method is able to select important and influential features from large data sets and plays a key role in increasing the efficiency and reducing the computational complexity of models.

2- Improving the accuracy and efficiency of learning models:

Using selected features in advanced classification algorithm models, such as random forest, to increase the prediction accuracy, reduce the error rate, and improve the overall performance of models, while minimizing the number of features used and computational complexity.

Research Questions

1- Can the hybrid algorithm of particle swarm optimization and genetic algorithm select an optimal and effective subset of features to improve the accuracy of learning models?

2- Does the use of features selected by the proposed algorithm increase the accuracy of advanced classification algorithm models such as decision forest compared to traditional feature selection methods?

3- Which features have the greatest impact on improving the performance of learning models and what is their role in increasing accuracy and reducing computational complexity?

Research hypotheses

1- The combined algorithm based on particle swarm optimization and genetic algorithm is able to identify an optimal and effective subset of features that leads to a significant improvement in the accuracy of machine learning models compared to using all features.

2- The use of features selected by the proposed algorithm significantly increases the accuracy of advanced classification algorithm models (such as random forest) compared to traditional feature selection methods.

3- The subset of features selected by the proposed algorithm significantly reduces the computational complexity (in terms of training time and number of features) without a noticeable decrease in the model's prediction performance.

4- Some features have a more significant impact on the performance of learning models than others, and removing unimportant or noisy features increases the stability and generalizability of the model.

Methodology

Structure and steps of the proposed method

As stated in the introduction, in this thesis, in order to improve the feature selection process and increase the accuracy of machine learning models, a hybrid method based on genetic algorithm and particle swarm optimization algorithm is used. The main goal of using this hybrid approach is to simultaneously utilize the fast search capability of particle swarm optimization algorithm and the ability of genetic algorithm to maintain population diversity and perform extensive search in the response space.

In the feature selection problem, the search space includes all possible subsets of features in the dataset, which grows exponentially with increasing number of features. In such circumstances, the use of traditional search methods is practically not possible and finding an optimal subset of features becomes a complex optimization problem. For this reason, the use of metaheuristic algorithms has been considered as an effective solution for searching in this large space. Combining metaheuristic algorithms can make the search process more efficient and increase the probability of achieving more optimal solutions by taking advantage of the advantages of each of them.

In the proposed method of this research, the feature selection process is carried out using a combination of genetic algorithms and particle swarm optimization. In this framework, each solution represents a subset of features and the quality of each subset is evaluated using the performance of a classification algorithm model. In this research, the random forest algorithm is used as a classification algorithm model to measure the quality of the selected features. In this way, the combined algorithm tries to identify a subset of features that can provide the highest accuracy of the classification algorithm.

In the following, Figure 3-1 shows the general diagram of the proposed method in this research. This diagram shows the different stages of implementing the proposed method, from data entry, preprocessing, feature selection process using the PSO–GA hybrid algorithm, to the final classification stage and model performance evaluation. In the rest of this chapter, each of the stages in this diagram will be examined and explained in detail.

1- Data Preprocessing

Data preprocessing is one of the important and fundamental steps in the data mining and machine learning process. In many real datasets, raw data usually has problems such as missing values, different scales between features, or noise. The presence of such problems can affect the performance of machine learning algorithms and reduce the accuracy and efficiency of models. Therefore, data preprocessing is performed with the aim of preparing data and improving their quality before entering the modeling stage. Performing this step correctly can play an important role in increasing the accuracy of learning models and improving the feature selection process.

In this study, before implementing the feature selection algorithm and the classification process, the datasets undergo several preprocessing steps to put the data in a format suitable for analysis and learning. The most important preprocessing steps in this research include missing value management, data normalization, and data partitioning into training and test sets.

One of the common problems in real datasets is the presence of missing or non-existent values in some features. The presence of these values can disrupt the model learning process. In this research, the three nearest neighbors method with a K value of 3 is used to replace missing values. In this method, for each sample with a missing value, the closest samples are identified based on the distance criterion in the feature space, and the missing value is estimated based

on the average of the same feature values among the three nearest neighbors. This method provides more accurate performance than many simple methods such as replacing with the average of the entire data, due to considering the similarity between samples.

After replacing the missing values, the data normalization step is performed. Since the features in the dataset may have different scales and ranges, this can cause some features to have a greater impact on the learning process. To avoid this problem and create a uniform scale among the features, this research uses the min-max normalization method. In this method, the values of each feature are converted to a specific range, usually between zero and one. The min-max normalization formula is expressed as follows:

$$X_{norm} = \frac{X - X_{min}}{X_{max} - X_{min}} \qquad (1-3)$$

In this regard, x is the initial value of the feature, X_{min} and X_{max} are the minimum and maximum values of that feature in the dataset, respectively, and X_{norm} is the normalized value. Using this method causes all features to be in a common range, and as a result, the learning and search process in optimization algorithms is performed with greater accuracy and stability.

Next, after completing the above steps, the data is divided into two parts: the training set and the test set. The purpose of this division is to train the machine learning model on a part of the data and evaluate its performance on data that has not been used in the training process. This allows the model performance to be evaluated more realistically and prevents overfitting. In this research, the main part of the data is considered for model training and a smaller part for testing and evaluating the model performance.

In summary, performing preprocessing steps including managing missing values using the three nearest neighbors method, normalizing data using the

Min–Max method, and dividing data into training and testing sets increases data quality and provides suitable conditions for implementing the feature selection algorithm and classification algorithm model in the later stages of the research. Figure 3-2 summarizes the preprocessing process used in this work.

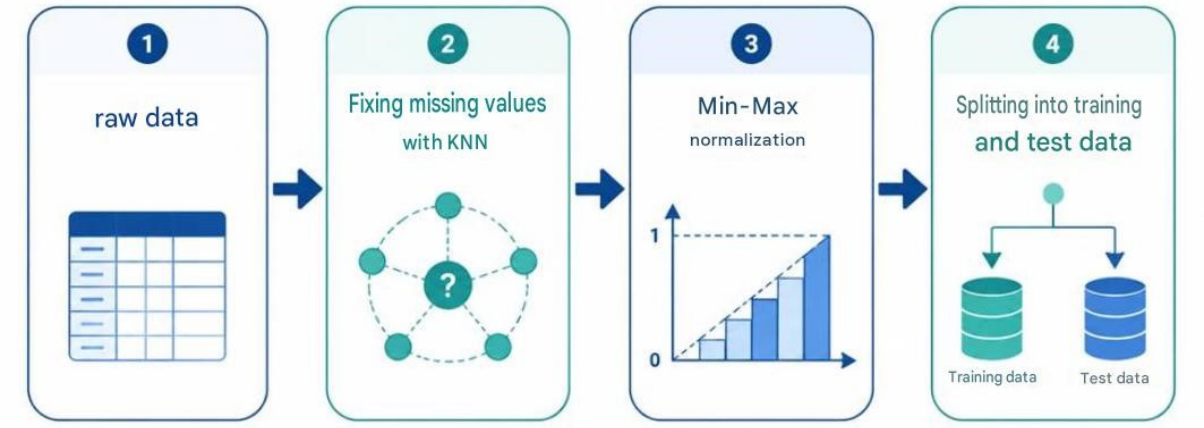


Figure 3-2 Summary of the preprocessing process

2- Feature selection using the combined process of genetic algorithm and particle swarm optimizer

In this research, the feature selection process is designed based on a combined structure of particle swarm optimization algorithm and genetic algorithm. The purpose of designing this structure is to create a dynamic balance between exploration in a very large search space of subsets of features and exploitation of promising areas to ultimately select a smaller but more effective set of features for the classification algorithm model. In this algorithm, each solution is represented as a binary chromosome with a length equal to the total number of features, where the value 1 indicates the selection of that feature and the value 0 indicates its elimination. Therefore, the search space will be equal to 2^n

(for n features), in which direct search is practically impossible and requires an effective metaheuristic method.

The combined algorithm used operates sequentially; This means that first the particle swarm optimizer-based update is performed and then genetic operators including crossover and adaptive mutation are applied. In the particle swarm optimizer stage, each particle has a binary position and a real velocity vector. The velocity update is performed according to the following classical relation:

$$v_i^{t+1} = wv_i^t + c_1 r_1 (gBest - x_i^t) + c_2 r_2 (gBest - x_i^t) \qquad (3\text{–}2)$$

where w is the inertia weight and decreases from 0.9 at the beginning of the algorithm to 0.4 at the end of the execution so that the algorithm does more exploration in the initial stages and has faster convergence in the final stages. The two parameters c1 and c2 are acceleration coefficients and both values are taken as 2. The vectors r1 and r2 are also vectors of uniform random numbers in the interval [0,1]. Since the search space is binary, the true velocity is converted into the probability of selecting each feature after calculation through the sigmoid function:

$$s(v) = \frac{1}{1+e^{-v}} \qquad (3\text{–}3)$$

And the final value of each bit is determined as follows:

$$x_{i,j}^{t+1} = \begin{cases} 1, & rand < s(v_{i,j}) \\ 0, & o.w \end{cases} \qquad (3\text{–}4)$$

In this way, the particle swarm optimization algorithm is responsible for guiding the population towards the best areas discovered so far, causing the algorithm to move towards high-quality solutions at an appropriate speed. After the particle swarm optimization stage, the genetic algorithm comes into play to increase the diversity of the population. In this stage, the two-point crossover operator is first performed. In this method, two parents are selected and two random points between 2 and nVar−1 are determined as cut points, and the middle part of these

two parents is exchanged with each other. This type of crossover allows the combination of meaningful blocks of features and the transfer of appropriate patterns from the parents to the next generation. After that, the mutation operator is applied adaptively and with a decreasing probability from 0.20 at the beginning of the algorithm to 0.02 at the end. The mutation is performed as a bit value inversion and plays an important role in preventing the algorithm from getting stuck in local optima. The gradual decrease in the mutation rate causes the algorithm to have a wide search at the beginning and to show convergent behavior at the end.

After the implementation of the particle swarm optimization algorithm, crossover and mutation, the elite mechanism is used to transfer the best individuals of the current generation to the next generation without change. This prevents high-quality solutions from being eliminated due to random operators. In addition, an external archive is also maintained that records the best answers throughout the entire process and prevents the accumulation of similar solutions.

The core of the algorithm is its fitness function, which evaluates the quality of each subset of features. In this research, the fitness function is defined based on the performance of the random forest classification algorithm and the simplicity of the model (the number of selected features). Initially, if no features are selected, the fitness value is set to 1 to eliminate such answers from the algorithm. In the at least one feature selection mode, the data is first subsampled based on the selected features and then 5-part cross-validation is performed. In each part, the random forest classifier algorithm is trained with 100 trees and its accuracy is calculated on the test data. The average accuracy of the five parts is recorded as the final accuracy. Then, a penalty is applied to the number of features to guide the algorithm towards selecting smaller subsets. The cost function is defined as follows:

$$Cost = \alpha(1 - Accuracy) + (1 - \alpha)\frac{\sum\ (position)}{n} \qquad (3\text{–}5)$$

where α = 0.99. This value indicates that the algorithm focuses on maximizing the accuracy of the classification algorithm, but at the same time, the size of the subset of features is also taken into account in decision-making. The first term, (1-Accuracy), directly indicates the error of the classification algorithm, and the second term expresses the ratio of selected features to the total number of features, which is a kind of penalty for more complex models. In this way, the algorithm seeks to minimize the cost. In each iteration, the combined PSO–GA process first quickly identifies suitable areas through the particle swarm optimization algorithm, and then helps to diversify the population and prevent premature convergence by using genetic operators. This structure makes the algorithm have both the fast optimization of particle swarm optimization and the power of genetic exploration. Finally, the best chromosome state that has the lowest cost function is introduced as the final subset of selected features. The following is a general diagram of the feature selection steps.

3- Classification algorithm using random forest

After an optimal subset of features was extracted in the previous step using the PSO GA hybrid algorithm, the next step is dedicated to training and evaluating the classification algorithm model based on this selected set of features. In this research, the random forest algorithm has been used to perform the classification algorithm process; a powerful and stable algorithm that, due to its ensemble structure, allows learning complex data patterns and effectively dealing with the overfitting phenomenon. Random forest is, in its essence, an ensemble learning method that is formed by putting together a large number of independent decision trees. Each of these trees is created based on a random sampling of the training data; a process known as Bagging. In this mechanism, to build each tree, a set of training examples is first selected with replacement

and then, at each node of the tree, instead of using all the features, only a random subset of the features is examined to find the best split. These two levels of randomization, namely sampling with replacement and random selection of features, reduce the correlation between the trees and increase the structural diversity of the model. The result of this process is a model whose output is the result of a set of diverse predictions and therefore has higher stability and accuracy.

The training process of the random forest classifier algorithm is carried out after the selection of features in such a way that the training data, including the examples and their correct labels, is presented to the model. Each decision tree in the forest forms its internal structure using the same data but with new random sampling. At each node, the algorithm determines the best possible split based on the selected features using criteria such as reducing uncertainty or increasing entropy. This process continues until stopping criteria such as reaching a certain depth are met. Given the independence of tree construction, the entire ensemble implicitly benefits from a form of parallel learning, which also leads to increased computational efficiency.

After the construction of all trees is completed, the trained model is used to predict the class of new examples. In the prediction phase, each decision tree independently proposes an output label based on its decision paths. Then, the random forest determines the final class by a majority voting mechanism among the predictions of all trees. This voting method ensures that the final decision is not affected by the possible error of a single tree, and as a result, the model output is more stable and reliable. For this reason, random forest usually provides more accurate and robust performance than a single decision tree.

One of the key points in using random forest in this research is that the model is trained solely on the training data after feature selection and no test data is used

to build the trees at any stage. This approach prevents data leakage and makes the evaluation results more reliable. Since the feature set is determined by the PSO GA algorithm and this algorithm also uses only the training data, the random forest classification algorithm is also trained based on the exact same extracted feature space and without the interference of future information.

It should also be noted that, unlike some progressive algorithms, random forest has a batch nature; this means that after the initial training, the structure of the trees remains fixed and if new data needs to be changed or added, the model must be retrained from the beginning. In this study, in order to properly evaluate the performance, the random forest model is retrained based only on the selected features of the same stage at each fitness calculation during the feature selection process. This repeated retraining ensures that the quality measurement of each feature subset reflects the actual behavior of the classification algorithm model in future conditions. Finally, the performance of the trained model on the test data is reported as the final measure of the accuracy of the classification algorithm.

Result

As explained in the research methodology chapter, the first step in implementing the proposed method was data preprocessing. In this step, the missing values in the data were identified and, in order to prevent disruption in the learning process, their values were assigned. For this purpose, the average of three nearest neighbors method was used; so that the value of each missing feature was estimated based on the average of the values of the same feature in three samples close to it. Using this method, the statistical structure of the data was preserved as much as possible and bias in the data was prevented.

Next, in order to equalize the scale of the features and prevent the unbalanced influence of features with larger amplitudes, Min–Max normalization was applied to the data. In this step, the values of all features were transferred to the interval [0,1] to provide more suitable conditions for training machine learning algorithms and to make the learning process more stable and faster.

After the preprocessing steps, the data were divided into two parts: training and testing. In this way, 80% of the data were used to train the model and the remaining 20% were used to evaluate the final performance of the model. This division allowed the model to be trained on the majority of the data, while its performance was evaluated independently on data that was not observed in the training process.

In the next step, in order to identify effective features and eliminate unnecessary features, a hybrid feature selection algorithm based on a combination of particle swarm optimization and genetic algorithm (PSO–GA) was implemented. At this stage, only the data related to the training part were provided to the algorithm so that the feature selection process could be carried out without using information from the test data.

To implement the feature selection algorithm, the algorithm parameters were adjusted empirically. In this study, the number of algorithm iterations was set to 20 and the population size was set to 20 members. Choosing these values allowed for a proper balance between search quality and computational time; So that the number of iterations is sufficient for the algorithm to reach the appropriate areas of the search space, while the execution time of the algorithm does not increase too much. Also, the inertia coefficient was chosen to be equal to 0.7 to create a good balance between exploration of the search space and exploitation of promising areas. The cognitive and social learning coefficients were set to be equal to 1.5, respectively, to consider the effect of the individual

experience of the particles as well as the best global response in a balanced way in the process of updating the positions. In addition, the mutation rate was chosen to be equal to 0.2 to prevent premature convergence of the algorithm by creating diversity in the population and to allow for the examination of more diverse combinations of features. In this algorithm, each response was represented as a binary vector of features; so that the value 1 indicated the selection of the feature and the value 0 indicated its non-selection. In each iteration, the particle positions were modified using the particle swarm algorithm update relations, and then, to increase the population diversity, the combination and mutation operators of the genetic algorithm were applied to some population members. This combination allowed the algorithm to benefit from the fast search ability of PSO to find suitable areas and from the population diversity of GA to avoid getting stuck in local optima.

The quality assessment of each feature subset was performed using the performance of the random forest classifier algorithm, the details of the objective function are provided in the research method chapter. At this stage, to accurately measure the performance of the feature subsets, a five-part cross-validation was performed, and in each part, the random forest classifier algorithm was trained with 100 decision trees. Finally, the average accuracy obtained from these steps was used as an evaluation criterion in the feature selection process.

In general, in the combined PSO–GA process, the particle swarm algorithm first quickly identified suitable areas of the search space, and then the genetic operators increased the population diversity by creating new combinations and applying mutations. This collaboration between the two algorithms made the search process more efficient and identified subsets of features that, while reducing the dimensionality of the data, also maintained the model's ability to perform classification.

Table 4-4 Parameters set in the PSO–GA feature selection algorithm

Parameter name	Symbol/Unit	Amount	Explanation
Number of iterations of the algorithm	Max Iteration	20	Number of times the optimization process is executed
Population size	Population Size	20	Number of particles/chromosomes per iteration
Inertia coefficient	w	0.7	Balancing global and local search in PSO
Coefficient of cognitive learning	c1	1.5	Effect of individual experience of each particle
Social learning coefficient	c2	1.5	Effect of best collective experience
Combination rate	Pc	0.8	Probability of performing the combination operator in the genetic algorithm
Mutation rate	Pm	0.2	Probability of random change of genes to maintain population diversity
Parent selection method	Selection Method	Roulette Wheel	Select individuals based on fitness value
Cross validation	K-Fold	5	Fairly evaluate the model and prevent overfitting
Combination operator type	Crossover Type	Single Point	Generate new offspring from selected parents
Type of mutation operator	Mutation Type	Bit Leap	Randomly change the value of some genes
elitism	Elitism	1 Member	Transfer the best response to the next generation without change
Feature selection objective function	Cost Function	Combination	Simultaneous minimization of the error of the random forest model classification algorithm and the number of selected features

After completing the optimization process and extracting the optimal subset of features by the PSO–GA hybrid algorithm, the final step of the research was dedicated to training and evaluating the classification algorithm model. At this stage, the Random Forest algorithm was used as a powerful ensemble learning model. In the configuration of the classification algorithm model, the following parameters were adjusted to achieve the highest efficiency:

1. Number of decision trees: The final model was built using 100 independent decision trees. Using this number of trees creates an optimal balance between the accuracy of the classification algorithm and the computational complexity and ensures that the variance of the model is minimized.

2. Aggregation method: The training process is based on the "Bagging" technique. In this method, each tree is trained on a random sample with replacement of the training data. This diversity in the input data of each tree makes the model as a whole insensitive to minor changes in the data set and has high generalizability.

3. Majority voting mechanism: The final class prediction is made through a voting process among all 100 trees. Each tree independently predicts a label and the class that receives the most votes is considered as the final output of the model. This approach neutralizes individual tree errors and significantly increases the final accuracy.

4. Node splitting criterion: In the structure of each tree, the Gini index was used to determine the best feature for splitting nodes to minimize data impurity at each stage.

After training the model on the selected data, its performance was measured using test data that did not play a role in the training process. In this step, the predicted labels were compared with the actual labels and the final evaluation criteria were calculated by extracting statistical and probabilistic values. Next, we will examine the results obtained for the two datasets introduced in Section 4-2.

Evaluation of the method results on the spam email dataset

Figure (4 1) shows the convergence trend of the hybrid PSO–GA algorithm for the feature selection process on the Spambase dataset. As is clear from this figure, the value of the cost function has decreased uniformly and stepwise

during 20 iterations of the algorithm, which indicates the efficiency of the hybrid search mechanism in gradually moving towards the space of more optimal responses.

In the initial iterations, the cost value is around 2.2* 〖10〗 ^(-3), which represents a combination of the classification algorithm error and the relatively large number of features. With the beginning of the search process, relatively large jumps in the cost value are observed; this behavior is due to the dominant role of the genetic component (Crossover and Mutation) in creating high diversity in the initial population and rapid exploration of different parts of the search space.

In the middle of the convergence trend, the cost reduction continues with a gentler slope. This section indicates the equilibrium stage in the hybrid algorithm; Where the PSO mechanism, using the collective knowledge and individual experience of the particles, is directed towards regions with more stable responses. This behavior is fully consistent with the nature of metaheuristic algorithms, which have a broad search in the initial phases and a more precise local search in the final phases.

Finally, after about 17 iterations, the cost value stabilizes in the range of $1*10^{(-3)}$ and does not change noticeably until the end of the iterations. This final stability indicates that the algorithm has reached a point where a balance between classification accuracy and the number of selected features has been achieved and further optimization has no noticeable effect on the cost function. The cessation of changes in the final few iterations indicates that the population has converged towards a relatively optimal solution and the algorithm is not trapped in local deadlocks or random fluctuations.

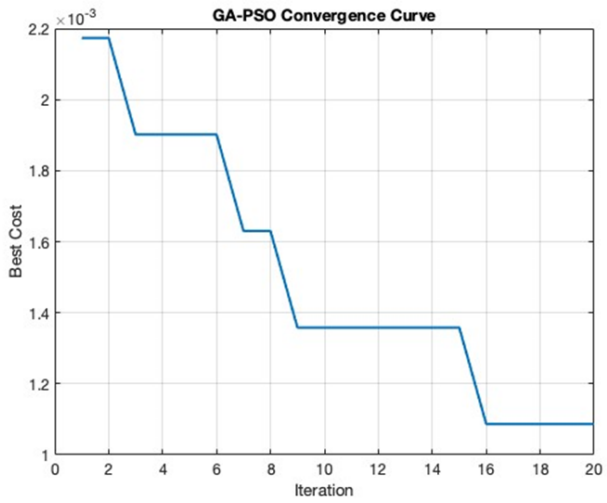


Figure 4 1: Convergence process of the hybrid PSO–GA algorithm for feature selection process on the Spambase dataset

At the end of the optimization process, the hybrid PSO–GA algorithm identified and selected 46 features from the initial 57 features of the Spambase dataset as more effective and important features. Then, using these features, the random forest classifier algorithm was trained and its performance was evaluated on the test dataset. Next, the results are examined and analyzed based on the confusion matrix and the receiver operating characteristic curve for the test dataset.

Figure 4 2 shows the confusion matrix of the results of the random forest classifier algorithm on the test dataset. In this study, class 0 represents healthy (non-spam) emails and class 1 represents spam emails. The confusion matrix is a tool for evaluating the performance of a classification algorithm model, which displays the number of correct and incorrect predictions for each class by comparing the actual and predicted labels.

Based on Figure 4 2, the elements of the confusion matrix are interpreted as follows:

• 538 healthy email samples (class 0) are correctly identified as healthy.

• 20 healthy email samples are incorrectly classified as spam.

• 23 spam email samples (class 1) are incorrectly identified as healthy.

• 339 spam email samples are correctly classified as spam.

These results indicate that the proposed model is able to discriminate between healthy and spam emails with reasonable accuracy, and the number of false positives, especially in identifying spam emails, is at an acceptable level.

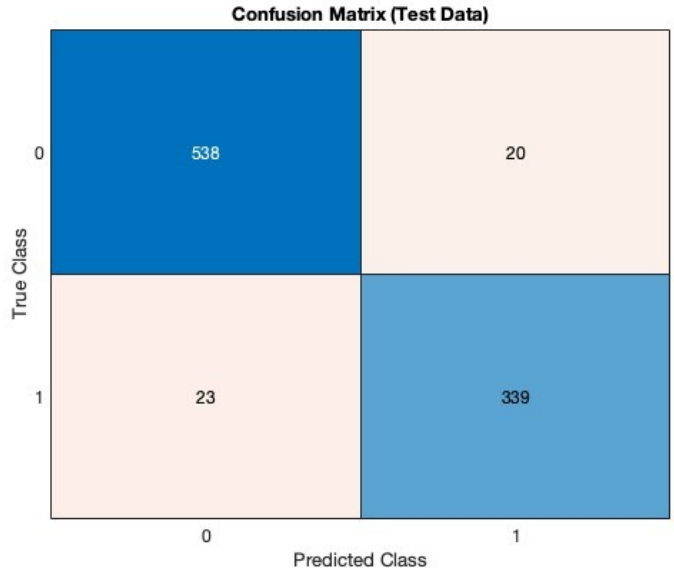


Figure 4 2: Confusion matrix of the results of the random forest classifier algorithm on the Spambase dataset

Next, Figure 4 3 shows the receiver operating characteristic (ROC) curve obtained for the Spambase test dataset. The ROC curve shows the relationship between the true positive rate (TPR) and the false positive rate (FPR) at different decision thresholds, thereby evaluating the ability of the model to distinguish between the two classes of healthy (0) and spam (1). In this graph, the dotted diagonal line (from (0,0) to (1,1)) represents the performance of a random classifier; the curve lying on this line means that the model has no significant discrimination power and its performance is approximately equivalent to random guessing. In contrast, the area above this line indicates better than chance performance and good discrimination power; The closer the curve is to the upper-left corner, the model with a lower FPR can achieve a

higher TPR, which is the most desirable case for spam detection problems. On the other hand, a curve below the diagonal line usually means poorer performance than chance and can indicate that the decision direction of the model is reversed (in such a way that performance is improved by reversing the outputs).

According to Figure 43, the ROC curve is clearly located above the diagonal line and most of its path is concentrated near the upper-left corner; this indicates that the random forest classifier (with features selected by PSO–GA) is able to correctly identify spam emails at a high rate, while the false positive error rate (falsely labeling healthy emails as spam) remains low. The area under the curve or AUC in this figure is also reported to be 0.988. An AUC value close to 1 indicates a very high discrimination power and the ability to correctly rank spam and healthy samples at most decision thresholds; in other words, the model assigns a higher score (or probability) to spam samples than healthy samples with a very high probability. Therefore, the location of the curve in the area above the reference line and the AUC value = 0.988 confirm that the proposed method has provided a very favorable and stable performance in terms of spam detection on the test data.

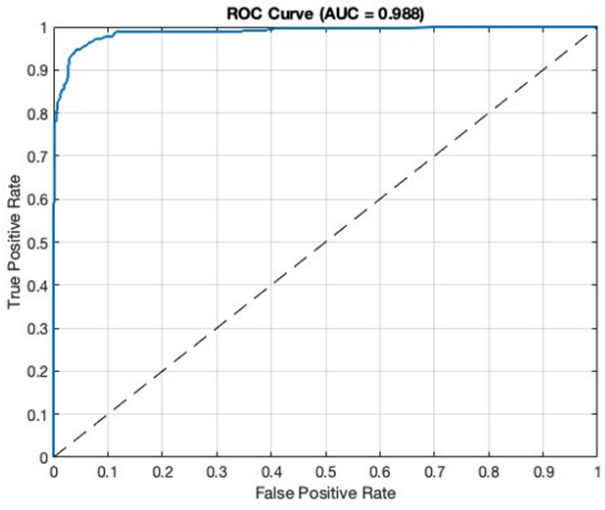


Figure 4-3: Receiver performance characteristic curve obtained for the Spambase test dataset

Figure 4-4 shows the performance evaluation results of the proposed model on the test dataset. According to this figure, the four main metrics including overall accuracy, spam label prediction accuracy, real spam recovery rate and AF score are reported to be 96.33%, 95.43%, 94.65% and 95.04%, respectively. The high value of overall accuracy indicates that the model was able to correctly classify a large proportion of healthy and spam emails. On the other hand, the high spam label prediction accuracy indicates that the majority of emails that the model detected as spam were actually spam; therefore, the false alarm rate is at a low level, which is of great importance from an application perspective, because incorrect labeling of healthy emails as spam can lead to the loss of important messages. Also, the acceptable value of the recovery rate shows that the model has a good ability to identify real spam and has detected a large part of them without being stuck; this is especially important for reducing false negative error and preventing spam from reaching the user's inbox. Finally, the value of the AFK score, which is the coordinated average of the prediction accuracy and recovery rate, confirms the appropriate balance between these two criteria and shows that the model has not simply improved one of the criteria at the expense of the other. In general, the closeness of the prediction accuracy, recovery rate, and AFK score values and their being all in a range higher than ninety-four percent indicates the stability of the model's performance and appropriate resolution on the test data; therefore, it can be concluded that the proposed method provides a good performance in a practical spam detection scenario by maintaining a balance between correct spam detection and reducing false alarms.

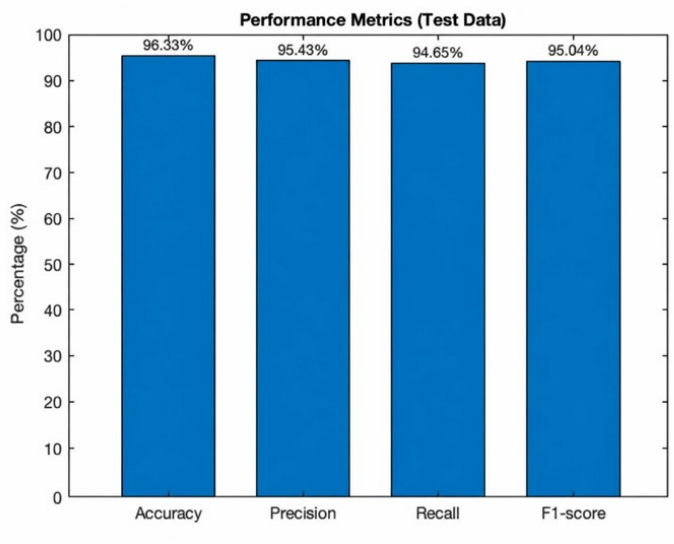


Figure 4-4 shows the results of evaluating the performance of the proposed model on the Spambase test dataset

• Review of the results of the method on the phishing dataset

Figure 4-5 shows the convergence trend of the combined method used on the phishing dataset. As can be seen in the graph, the value of the cost function has decreased with a steep slope in the initial iterations; this rapid drop indicates that the algorithm has succeeded in eliminating a significant portion of inappropriate feature combinations in the very early stages and moving towards more promising areas of the search space. After this stage, the slope of the cost reduction becomes gentler and the graph gradually approaches a stable region; so that from about the ninth iteration onwards, the cost changes are very limited, indicating that the algorithm has reached a converged state and further improvements are only possible on a very small scale. This behavior indicates that the search process has reached a good balance between broad search in the early stages and focused search in the late stages, and aimless fluctuations and getting stuck in shallow spots have been prevented. Finally, the final result of running the algorithm on the phishing dataset shows that out of the initial thirty features, twenty-two features have been selected as the superior feature set; in addition to reducing the data volume and simplifying the model, the selection of

this optimal subset has increased the processing speed and focused the model on the features that are truly influential in distinguishing healthy and phishing samples.

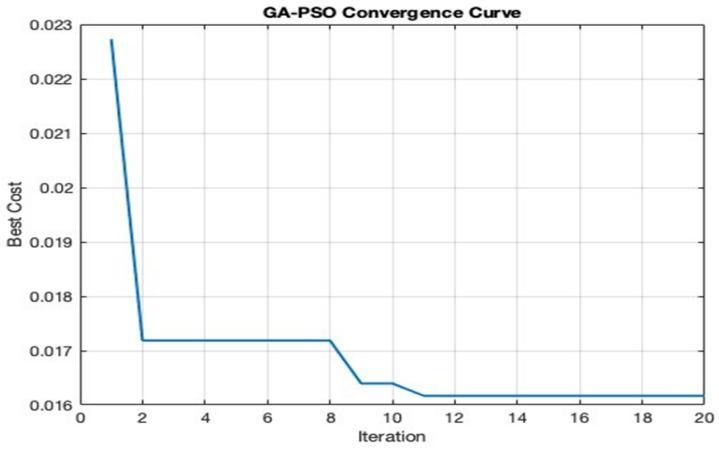


Figure 4-5: Convergence trend of the hybrid PSO–GA algorithm for feature selection process in the phishing dataset

Figure 4-6 shows the confusion matrix of the phishing dataset classification results in the test phase. In this matrix, label -1 indicates healthy websites and label 1 indicates phishing websites. Based on the obtained values, 955 healthy samples were correctly classified in the healthy class, while 25 healthy samples were incorrectly classified in the phishing class. The relatively low value of this error indicates that the model performed reasonably well in detecting healthy pages and was able to prevent false alarms to a large extent.

On the other side of the matrix, among the phishing samples in the test set, 1,216 samples were correctly identified as phishing, indicating the high power of the model in detecting malicious pages. In contrast, only 15 phishing samples were incorrectly identified as healthy, which is a very small number compared to the total number of phishing samples. The low value of this value indicates that the proposed model has a good ability to identify phishing threats and the probability of malicious pages passing the detection system has been significantly reduced.

In general, comparing the values in the matrix elements shows that the number of correct predictions in both classes is much higher than the classification errors. This indicates that the proposed model has been able to identify the effective features associated with healthy and phishing pages well and distinguish between these two categories with high accuracy.

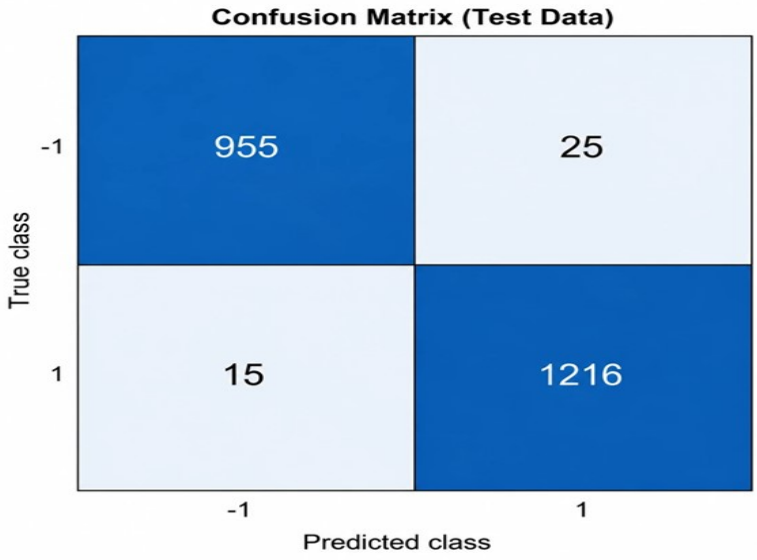


Figure 4-6: Confusion matrix for the phishing dataset classification results in the test phase

Figure 4-7 shows the ROC curve for the phishing dataset. As can be seen from the graph, the area under the curve is 0.996, indicating that the model is almost perfectly capable of distinguishing between healthy and phishing websites. The curve increases very steeply and almost vertically at the very beginning of the axis, reaching the upper regions of the graph; this indicates that the model is able to correctly identify a significant portion of phishing samples while the false positive error rate is still almost zero. The curve then remains close to unity for almost the entire length of its path, indicating that the model maintains a high rate of correct identification of phishing samples. The large distance of the curve from the diagonal line, which represents random performance, indicates that the model is able to distinguish between the two classes with high confidence and is not confused. Overall, the shape of the curve and the high

AUC value make it clear that the model has a very accurate and reliable performance in balancing the correct detection rate and controlling the classification error, and has been able to provide near-ideal performance in an application such as phishing detection, where both sensitivity and specificity are crucial.

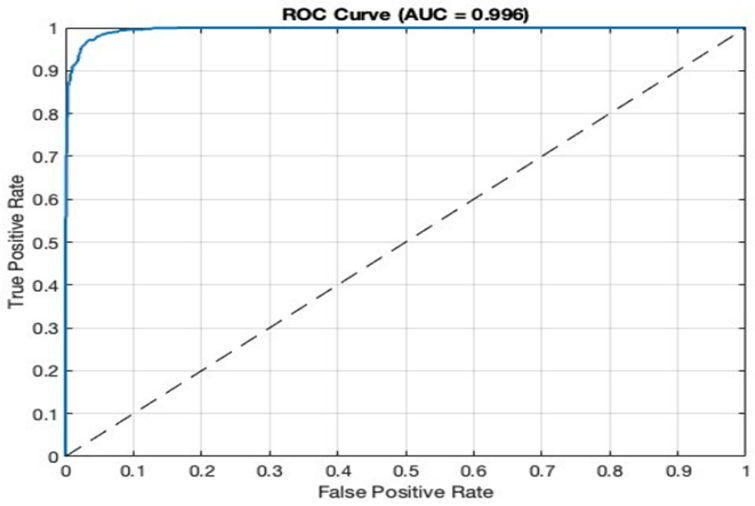


Figure 4-7: Receiver operating characteristic curve obtained for the phishing test dataset

Figure 4-8 shows the four main criteria for evaluating the model's performance in the testing phase. As can be seen from the graph, the model's accuracy value is 98.20%, which indicates that the model was able to correctly classify a very large portion of healthy and phishing samples. The accuracy value is also reported to be 97.98%, which indicates that when the model identifies a sample as phishing, this decision is almost always correct and the false positive error rate remains very low. The model's recall also reached 98.80%, which indicates its very high ability to identify phishing samples and indicates that only a very limited number of malicious samples are hidden from the system's view. Finally, the FEC score, which balances precision and recall, was 98.39%, indicating that the model performed consistently and consistently in simultaneously reducing false positives and false negatives. The sum of these values confirms that the model used performed at a very desirable level in terms

of overall accuracy, threat detection ability, and false alarm prevention, and is fully reliable for sensitive security applications such as phishing detection.

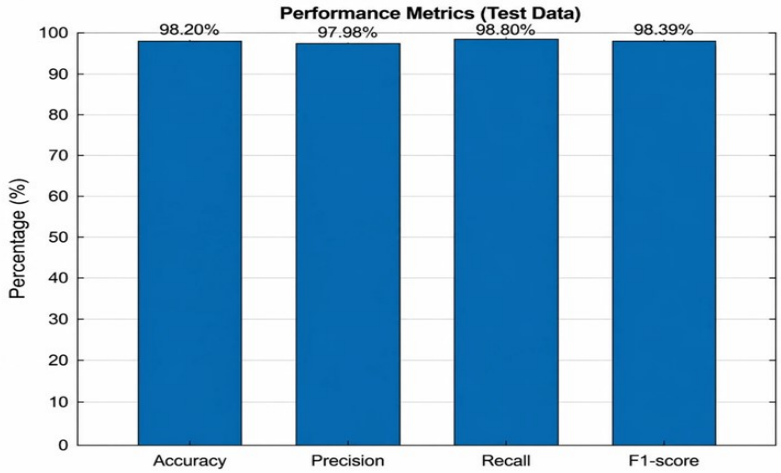


Figure 4-8: Results of evaluating the performance of the proposed model on the phishing test dataset

Conclusion

In recent years, the increase in data volume and number of features in machine learning problems has made feature selection one of the most important steps in data preprocessing. The presence of unnecessary and repetitive features can increase computational complexity, reduce processing speed, and degrade the performance of classification models. Therefore, using optimal methods to select effective features plays an important role in improving the accuracy and efficiency of machine learning models. In this study, a hybrid method based on PSO and GA metaheuristic algorithms was presented to select optimal features, and its performance was evaluated using the random forest classifier algorithm.

In the proposed method, first, the PSO GA hybrid algorithm was used to search the feature space and find an optimal subset of effective features. The simultaneous use of the global search capability of the PSO algorithm and the ability to generate diverse responses in the genetic algorithm made the feature

selection process more accurate and stable. Then, the selected features were given to the random forest classifier algorithm to examine the performance of the model in terms of classification accuracy.

Two datasets, Spambase and Phishing, were used to evaluate the proposed method. The results showed that the combined algorithm was able to remove a significant number of unnecessary features while maintaining or improving the classification accuracy. In the Spambase dataset, 46 features were selected from among the 57 initial features, and the model was able to achieve an accuracy of 96.79%, a precision of 96.40%, a recall of 97.89%, and an F1 score of 97.14%. Also, the AUC value of 0.988 indicated the high power of the model in separating classes.

In the Phishing dataset, the proposed algorithm selected 22 effective features from among the 30 initial features. The classification results showed that the model achieved an accuracy of 98.20%, a precision of 97.98%, a recall of 98.80%, and an F1 score of 98.39%. The AUC value of 0.996 also showed that the selected features have a very high ability to separate classes.

In general, the results of this study showed that the combined PSO GA method is an effective approach for feature selection in classification problems and can improve the performance of the model while reducing the number of features. The use of selected features reduced the complexity of the model, increased the processing speed, and improved the generalizability of the classification algorithm. Also, the stability of the results in two different datasets indicates the reliability and appropriate efficiency of the proposed method in feature selection problems.

Finally, it can be stated that feature selection using meta-heuristic algorithms, especially hybrid methods, can play an important role in increasing the efficiency of machine learning models, and the method presented in this research has also been able to provide acceptable results in this field.